Marker Signal Void: Linguistic Deception Fingerprinting and Transformer Benchmarking on a Domain-Stratified Bangla Rumour Corpus

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Abstract

Rumour detection in Bangla remains severely under-resourced, with prior work conflating rumour detection with fake news classification, absent domain stratification, and no principled analysis of what linguistic properties distinguish misinformation from authentic content in this language. We introduce MisInfo-11K, a domain-stratified corpus of 11,198 instances spanning seven topical domains — political, health, religious, cultural, sports, celebrity, and international — constructed through systematic web scraping of Bangla fact-checking portals and verified news outlets. We design 14 linguistically motivated markers and demonstrate through chi-square and Kruskal-Wallis analysis that 13 of 14 exhibit statistically significant cross-domain variation, revealing distinct deception strategies: political rumours mimic journalistic authority through elevated claim-verb density, religious rumours deploy emotional provocation via alarm signalling, and health rumours amplify fear through co-occurring medical claims and clickbait patterns. Building on this, we introduce the Marker Signal Void (MSV), a chi-square-weighted scoring framework that identifies rumour instances carrying no net discriminative surface signal relative to the non-rumour baseline, establishing a theoretical detection ceiling of Macro-F1 = 0.9843 for any surface-level approach. We benchmark three pretrained transformers — BanglaBERT, MuRIL, and XLM-RoBERTa — alongside five traditional models. MuRIL achieves the strongest performance at Macro-F1 0.9800, while a Random Forest augmented with our linguistic markers reaches 0.9750, surpassing XLM-RoBERTa. The gap between the MSV ceiling and transformer performance quantifies the role of contextual semantic representations in Bangla rumour detection beyond what surface-level linguistic cues can achieve.

1 Introduction

The 2019 Padma Bridge lynchings — in which crowds killed several people acting on a false Facebook rumour that children's heads were needed as sacrificial offerings for the bridge's construction — illustrate that unverified claims in active circulation cause physical harm well before their veracity can be established (Shirina and Prodhan, 2020). Yet computational approaches to Bangla *rumour detection* remain absent from the literature. Existing work addresses fake news detection — binary classification of fabricated articles — which is categorically distinct from rumour detection: rumours are unverified claims whose veracity is actively contested at propagation time (Zubiaga et al., 2018). No publicly available Bangla rumour corpus exists, no study has characterised linguistic properties of Bangla rumours across domains, and no theoretical framework has examined the limits of surface-level detection in this language. We introduce **MisInfo-11K** through four contributions: (**i**) the first domain-stratified Bangla rumour corpus spanning seven topical domains, establishing the empirical foundation for cross-domain analysis; (**ii**) 14 validated linguistic markers producing domain-specific deception fingerprints that reveal what properties distinguish Bangla rumours from authentic content, via χ^2 and Kruskal-Wallis analysis; (**iii**) the **Marker Signal Void (MSV)**, a novel chi-square-weighted scoring framework establishing Macro-F1 = 0.9843 as the theoretical upper bound for any surface-level approach; (**iv**) a benchmark of three pretrained transformers and five traditional models quantifying whether contextual representations contribute signal beyond surface markers — MuRIL achieves Macro-F1 0.9800 and a marker-augmented Random Forest reaches 0.9750, surpassing XLM-RoBERTa.

2 Related Work

Rumour Detection. PHEME (Zubiaga et al., 2016) established rumour detection with Twitter conversations across five breaking news events; RumourEval (Gorrell et al., 2019) formalised stance and veracity subtasks. Bi-GCN (Bian et al., 2020) demonstrated that propagation structure carries discriminative signal beyond text. A consistent finding is that linguistic features degrade across event domains (Zubiaga et al., 2018), directly motivating our domain-stratified analysis.

Bangla Fake News Detection. Hossain et al. (2020) introduced BanFakeNews (≈50K), the foundational Bangla benchmark with traditional and neural baselines. Subsequent work has applied CNN-LSTM, bidirectional GRU, and BERT-based classifiers to this corpus. All prior work is confined to fake news classification on a domain-undifferentiated corpus; none addresses rumour detection, domain stratification, or detection limits.

Linguistic Markers and Low-Resource Misinformation. Claim verbs, attribution patterns, and negation density have been shown discriminative for misinformation across languages(Newman et al., 2003), but these analyses are uniformly single-domain, masking cross-domain variability. Low-resource languages remain severely underserved(Chalehchaleh et al., 2024) and deception cues do not transfer reliably across morphologically distinct languages, reinforcing the need for language-specific corpus construction and marker analysis.

Pretrained Language Models for Bangla. (Conneau et al., 2020) introduced BanglaBERT, pretrained on 27.5 GB of Bangla web text, establishing state-of-the-art results across four Bangla NLU tasks. MuRIL (Khanuja et al., 2021) extends multilingual BERT with dedicated South Asian language support across 17 languages including Bangla. XLM-RoBERTa (Conneau et al., 2020) provides the standard high-capacity multilingual baseline. We evaluate all three on the first domain-stratified Bangla rumour corpus to establish reproducible baselines across model families.

3 Dataset: MisInfo-11K

3.1 Collection Methodology

Rumour instances were collected from three dedicated Bangla fact-checking portals — **Rumor Scanner**,[1] **Boom Bangladesh**,[2] and **Jachai**[3] — using a multi-threaded scraper dispatching 10 parallel worker threads with a 0.3-second inter-request delay. Rumor Scanner and Jachai were scraped via sequential article ID traversal; Boom Bangladesh via paginated listing crawls. URL-level deduplication within the training split prevented duplicate collection. Each fact-check headline is treated as a rumour instance (label 1).

Non-rumour instances were collected from nine established Bangladeshi news outlets — **Naya Diganta**, **NTV**, **MY TV**, **Manobzamin**, **RisingBD**, **BD Pratidin**, **Bangla Vision**, **Kaler Kantho**, and **Jugantor** — via RSS feeds and paginated HTML listings. Only headlines passing a Unicode Bangla character filter and a 15–300 character length constraint were retained. Non-rumour labels were assigned programmatically and never sourced from fact-check platforms to prevent label leakage.

3.2 Domain Taxonomy

Seven domains were defined a priori based on the thematic coverage of Bangla fact-checking portals: political, health, religious, cultural, sports, celebrity, and international. Labels were assigned automatically via a Bangla keyword lexicon curated per category. Rumour instances may carry multiple domain labels reflecting co-occurring themes; non-rumour instances carry exactly one label derived from the source category. Instances with no keyword match default to political, reflecting political content's empirical dominance in Bangla misinformation and its broad topical overlap with unlabelled instances.

3.3 Dataset Statistics

MisInfo-11K contains 11,198 instances across training (8,198), validation (1,400), and test (1,400) splits, each evaluation split balanced 50/50 between classes. Table 1 reports the full breakdown by split. Figure 2 shows domain distribution across both classes, revealing a notable asymmetry: political is the dominant rumour domain (2,407 instances) yet the most underrepresented in non-rumour (204), while sports and religious show the inverse pattern. This asymmetry motivates the domain-stratified analysis in Section 4.2.

[1] https://rumorscanner.com/
[2] https://boombd.com
[3] https://jachai.org

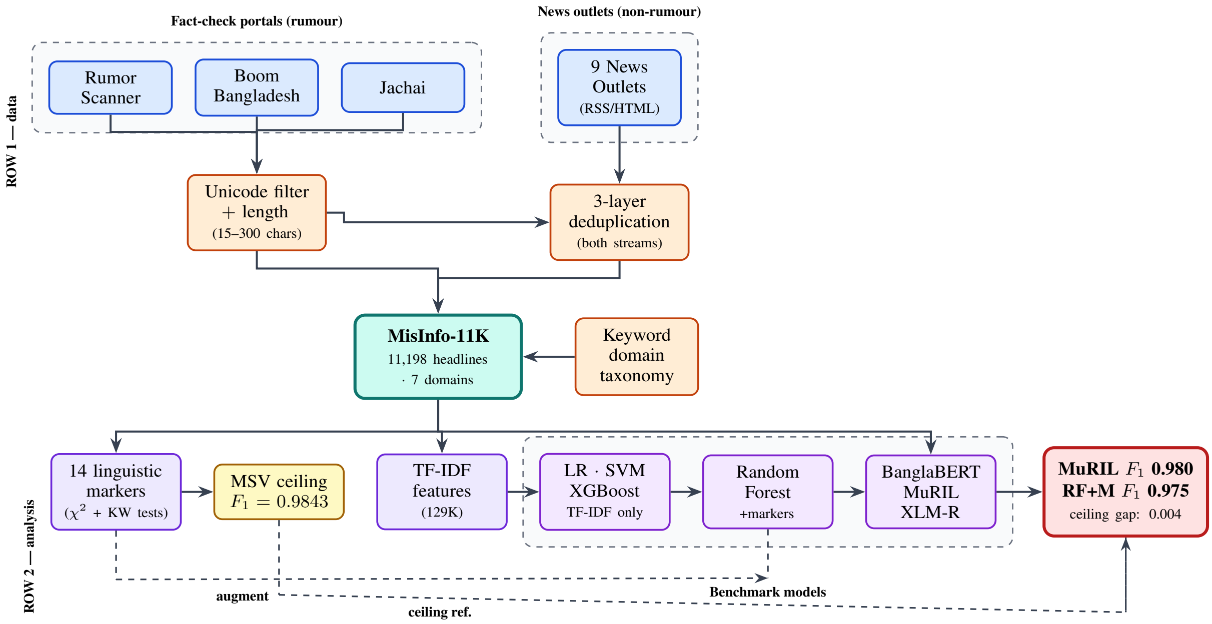

Figure 1: **MisInfo-11K pipeline. Row 1:** Rumour headlines from three fact-checking portals and non-rumour headlines from nine news outlets pass through quality-control filters and 3-layer deduplication (both streams) into the domain-stratified corpus. **Row 2:** Three parallel paths — (*i*) linguistic markers → MSV ceiling F_1=0.9843; (*ii*) TF-IDF → traditional classifiers with marker augmentation for Random Forest; (*iii*) transformer fine-tuning — converging on the benchmark results.

3.4 Quality Control

Three layers of quality control were applied. First, a URL and headline blacklist built from the training split prevented any training instance from appearing in validation or test. Second, exact-match deduplication on the `text` field removed conflicting labels arising from the same headline appearing across multiple sources. Third, explicit cross-contamination checks confirmed zero overlap across all three splits after cleaning. Instances shorter than five characters, containing only ASCII, or matching known portal homepage titles were discarded. The final corpus contains no duplicate headlines and no cross-split label leakage.

Beyond deduplication, taxonomy reliability was validated through human annotation. Three annotators independently labelled a 200-instance stratified subset across all seven domains. Pairwise Cohen's κ was computed for each domain, yielding a macro-averaged κ of 0.746 (*good* agreement; Landis and Koch 1977). Agreement was strongest for health ($\kappa = 0.861$) and sports ($\kappa = 0.847$), domains with unambiguous topical boundaries. Gold-standard labels for this subset were assigned by majority vote across the three annotators, with domain-level findings interpreted accordingly.

Split	**Total**	**Rumour**	**Non-R**	**Balanced**
Train	8,198	3,998	4,200	✗
Validation	1,400	700	700	✓
Test	1,400	700	700	✓
Total	**11,198**	**5,398**	**5,800**	—

Table 1: MisInfo-11K dataset statistics by split.

4 Linguistic Marker Framework

4.1 Marker Design and Validation

We design 14 linguistically motivated markers spanning four categories: *structural* (text length, word count, short title, long title, ends with question mark), *lexical* (claim verb, negation words, clickbait), *attribution* (has quote, unnamed source, named outlet), and *content* (has numbers, religious alarm, health claim). Markers were defined using Bangla-specific regular expressions validated against native speaker intuition and prior cross-lingual deception detection literature (Mahyoob et al., 2021; Zhou and Zafarani, 2020).

Each marker was validated on the full combined dataset using chi-square tests for binary markers and Mann–Whitney U tests for continuous markers ($p < 0.0001$ for all 14). Table 2 reports discriminative weights $w_m = \chi^2_m / \sum \chi^2$ and per-class activation rates. For the two continuous markers (word count and text length), the χ^2 column reports

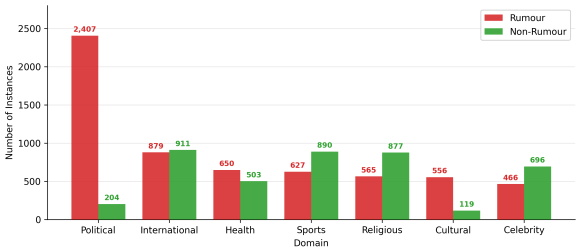

Figure 2: Domain distribution across rumour and non-rumour instances in the training set. Political is the dominant rumour domain (2,407 instances) yet the most underrepresented in non-rumour (204), revealing a structural asymmetry that motivates domain-stratified evaluation. Sports and religious show the inverse pattern.

a variance-normalised effect size used solely for weight computation, while significance was measured using Mann–Whitney U tests.[4]

Figure 3 visualises binary marker activation rates and the text length distribution by class.

Marker	χ^2	w_m	R	NR
Word count†	2881.07	0.311	9.83	7.30
Text length†	2494.79	0.269	65.80	49.55
Claim verb	1582.03	0.171	28.5%	1.6%
Long title	907.93	0.098	17.3%	0.9%
Has quote	287.90	0.031	5.4%	0.1%
Has numbers	254.17	0.027	10.2%	21.3%
Short title	237.01	0.026	1.2%	7.1%
Named outlet	151.66	0.016	2.8%	0.0%
Health claim	141.76	0.015	0.7%	4.2%
Negation words	87.59	0.009	16.0%	10.0%
Ends question	86.00	0.009	1.0%	3.8%
Unnamed source	64.43	0.007	1.8%	0.2%
Clickbait	50.87	0.006	1.4%	0.2%
Religious alarm	49.75	0.005	2.7%	0.9%

Table 2: Linguistic marker discriminative weights. $w_m = \chi^2_m / \sum \chi^2$. R and NR denote mean activation rates (binary markers) or mean values (continuous markers, †) for rumour and non-rumour instances respectively.

4.2 Domain Deception Fingerprints

We compute the mean activation rate of each marker separately for rumours in each of the seven domains, producing a 7×14 deception strategy profile matrix. Kruskal–Wallis tests confirm that 13 of 14 markers show statistically significant cross-domain variation ($p < 0.05$). Only `negation_words` fails to reach significance ($H = 10.34$, $p = 0.111$), rejecting the hypothesis that Bangla rumours employ a uniform linguistic strategy across topics.

Figure 4 shows the normalised heatmap of marker activation rates per domain. Three broad deception patterns emerge. Political and international rumours show elevated `claim_verb` rates (0.260 and 0.307 respectively), consistent with an *authority mimicry* strategy that imitates credible reporting without credible attribution (Rashkin et al., 2017). Religious rumours show the highest `religious_alarm` rate (0.196), suggesting *emotional provocation* as the primary mechanism. Health rumours are distinguished by elevated `health_claim` and `has_numbers` co-occurrence, reflecting *fear amplification* through pseudo-statistical medical claims (Pérez-Rosas et al., 2018). Pairwise cosine similarities between domain profiles exceed 0.999 (range: 0.997–1.000),[5] confirming that while domains differ in marker emphasis, they share a common underlying structure disrupted by domain-specific signals.

5 Experiments

5.1 Models

We benchmark eight models across two families.

Traditional Models. Five models combine character n-gram TF-IDF (2–5-grams) and word n-gram TF-IDF (1–3-grams), yielding 129,199 features:

1. Logistic Regression + TF-IDF
2. LinearSVC + TF-IDF
3. XGBoost + TF-IDF
4. Random Forest + TF-IDF + Markers
5. XGBoost + TF-IDF + Markers

Models 1–3 test surface pattern detection alone, while Models 4–5 test whether our 14 validated markers add discriminative signal beyond n-gram features.

Transformer Models. Three pretrained transformers are fine-tuned using a `[CLS]` $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(768$\rightarrow$2) classification head:

- **BanglaBERT** (Bhattacharjee et al., 2022), pretrained on 27.5 GB of Bangla text.
- **MuRIL** (Khanuja et al., 2021), a multilingual model with dedicated South Asian language support pretrained on 17 Indian languages including Bangla.

[4] Mann–Whitney U statistics: text length $U = 22{,}564{,}338$; word count $U = 23{,}117{,}884$; both $p < 0.0001$.

[5] Computed over the $\binom{7}{2} = 21$ pairwise domain profile vectors.

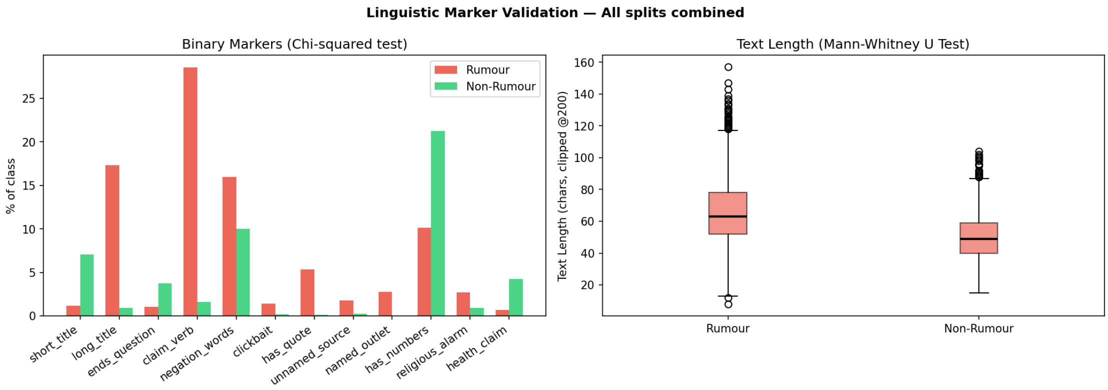

Figure 3: Left: binary marker activation rates by class (all χ^2, $p < 0.0001$). Right: text length distribution by class (Mann–Whitney U, $p < 0.0001$). Rumours are systematically longer and more claim-verb dense, while non-rumours cite numbers and ask questions more frequently.

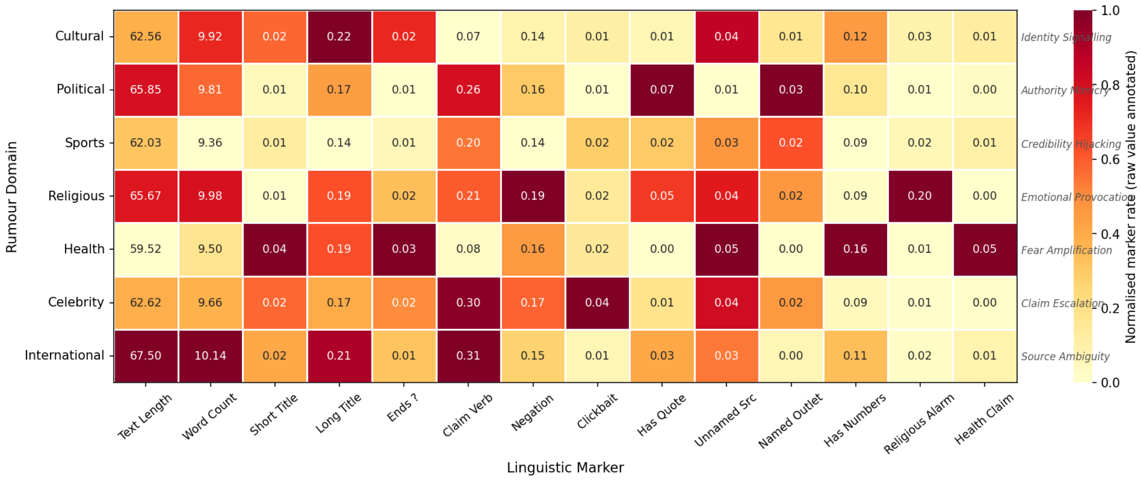

Figure 4: Domain deception strategy fingerprints. Colour intensity shows normalised marker activation rate, with raw values annotated. Strategy labels on the right axis reflect the dominant deception mechanism per domain.

- **XLM-RoBERTa** (Conneau et al., 2020), the standard 125M-parameter multilingual baseline.

5.2 Training Setup

All transformers were trained for 4 epochs using AdamW ($lr = 2 \times 10^{-5}$, weight decay = 0.01), linear warmup over 10% of steps, maximum sequence length of 128 tokens, and batch size 16.

Class-weighted cross-entropy loss and a weighted random sampler jointly address training-set imbalance (3,998 rumour vs. 4,200 non-rumour instances). Best checkpoint selection used validation Macro-F1, and all random seeds were fixed at 42.

Traditional models used validation-set grid search over $C \in \{0.01, 0.1, 1.0, 10.0\}$ for Logistic Regression and SVM. XGBoost and Random Forest used fixed robust defaults (200 estimators, maximum depth 6).

5.3 Results

Table 3 reports Macro-F1, ROC-AUC, and accuracy on the held-out test set for all eight models.

MuRIL achieves the highest test Macro-F1 of 0.9800 (validation: 0.9821), followed by BanglaBERT (0.9771; validation: 0.9807) and XLM-RoBERTa (0.9736; validation: 0.9750). All validation-to-test Macro-F1 drops remain within 0.004 points (MuRIL: 0.0021; BanglaBERT: 0.0036; XLM-RoBERTa: 0.0014), indicating consistent generalisation.

Among traditional models, Random Forest augmented with our linguistic markers (0.9750) surpasses XLM-RoBERTa (0.9736), confirming that hand-crafted markers carry genuine discriminative value beyond n-gram surface patterns. The +0.57 Macro-F1 gain of RF+Markers over the best TF-

IDF-only model (LR and XGBoost, both 0.9693) quantifies the independent contribution of the 14 validated markers.

Model	Macro-F1	ROC-AUC	Acc.
Traditional Models			
LR + TF-IDF	0.9693	0.9937	0.9693
SVM + TF-IDF	0.9643	0.9946	0.9643
XGBoost + TF-IDF	0.9693	0.9930	0.9693
RF + TF-IDF + M†	0.9750	0.9972	0.9750
XGB + TF-IDF + M†	0.9636	0.9934	0.9636
Transformer Models			
XLM-RoBERTa	0.9736	0.9937	0.9736
BanglaBERT	0.9771	0.9963	0.9771
MuRIL	**0.9800**	**0.9971**	**0.9800**

Table 3: Test-set results. † = marker-augmented. Best in bold.

5.4 Cross-Dataset Generalisation

To assess transferability beyond BanglaMisinfo-11K, we evaluate our best-checkpoint BanglaBERT model zero-shot on BanFakeNews (Hossain et al., 2020), a publicly available Bangla dataset comprising 7,000 authentic and 1,000 fake/satire news articles.

We sample 1,000 instances per class without any fine-tuning or domain adaptation, mapping BanFakeNews labels to our binary schema (authentic $\rightarrow$ non-rumour; fake/satire $\rightarrow$ rumour).

The model achieves Macro-F1 = 0.4728 and accuracy = 0.5325, with strong class asymmetry: recall on the fake class reaches 0.869, while authentic recall collapses to 0.196, producing 804 false positives against 131 false negatives.

This result is interpretable rather than anomalous, and arises from two compounding factors.

First, there is a severe input-length mismatch: BanglaMisinfo-11K consists of short claim headlines (mean 65.8 characters; truncated at 128 tokens), whereas BanFakeNews comprises full article bodies with mean length substantially exceeding this window. The model therefore operates on heavily truncated prefixes that fail to represent the article's discriminative content.

Second, the two tasks differ categorically: rumour detection concerns unverified claims in active social-media circulation, while fake-news detection concerns fabricated article content with a different linguistic register, source structure, and propagation context

Rather than indicating a generalisation failure, this result provides direct empirical support for the core motivation of BanglaMisinfo-11K: Bangla rumour detection requires a dedicated headline-level corpus and task formulation that existing full-article fake-news resources cannot substitute.

6 Analysis

6.1 Marker Signal Void

We introduce the **Weighted Marker Score** (WMS) for an instance x:

$$\mathrm{WMS}(x) = \sum_{m=1}^{14} w_m \cdot (f_m(x) - \mu_m^{\mathrm{NR}}) \quad (1)$$

where $f_m(x)$ is the value of marker m, $w_m = \chi_m^2 / \sum \chi^2$ is its normalised discriminative weight (Table 2), and μ_m^{NR} is the mean of marker m over the non-rumour training split, computed without access to evaluation data to prevent leakage.

An instance lies in the **Marker Signal Void** when $\mathrm{WMS}(x) \leq 0$: it carries no net positive marker signal relative to the non-rumour baseline, meaning surface-level linguistic cues are insufficient to distinguish it from authentic content.

We find that 3.1% of test-set rumours (22 of 700) lie in the void. Figure 5 shows the WMS distribution by class: the two distributions overlap substantially near zero, confirming that a non-trivial subset of rumours is surface-linguistically indistinguishable from authentic content regardless of model capacity.

Examining the 22 void instances reveals three recurrent patterns:

1. *Meta-referential misinformation*: headlines from fact-checking organisations reporting their own annual detection statistics (e.g., "472 pieces of misinformation identified in March"), which carry the register of credible journalism.
2. *Government scheme impersonation*: rumours written in neutral administrative language with no claim verbs, vague attribution, or clickbait markers.
3. *Recurring death hoaxes*: bare declarative statements where the rumour keyword itself appears in both genuine fact-check headlines and fabricated claims, providing no discriminative signal.

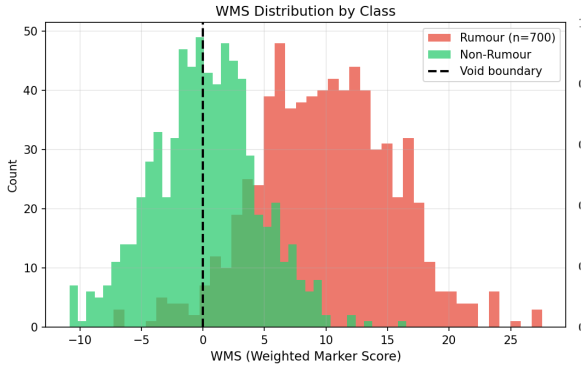

Figure 5: WMS distribution by class. The dashed line marks the void boundary (WMS $\leq$ 0); 22 rumours (3.1%) fall below it. Overlap near zero confirms a subset of rumours is linguistically indistinguishable from authentic content at the surface level.

6.2 Detection Ceiling and Transformer Contribution

The 3.1% void rate establishes a hard theoretical ceiling on any system relying exclusively on the 14 surface markers. A perfect oracle achieves at most Macro-F1$_{\text{ceil}}$ $= 0.9843$, corresponding to a theoretical recall ceiling of 0.9686 on the rumour class.

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All eight models operate below this ceiling. The ordering

TF-IDF only $<$ TF-IDF + Markers $<$ Transformer

holds consistently across the benchmark.

Explicit linguistic markers add +0.57 Macro-F1 points over distributional n-gram features (RF+Markers: 0.9750 vs. LR/XGB: 0.9693), while transformer pre-training adds a further +0.50 points over the best marker-augmented model (MuRIL: 0.9800 vs. RF+Markers: 0.9750).

Each component contributes independent, additive discriminative value.

MuRIL (FN = 10) recovers 12 of the 22 void rumours through contextual semantic representations; BanglaBERT (FN = 11) recovers 11; XLM-RoBERTa (FN = 14) recovers 8.

The 8–14 false negatives that persist across all three transformers constitute the irreducible difficulty of this dataset under text-only approaches (Zubiaga et al., 2018): rumours whose surface form is indistinguishable from authentic content even to contextual models.

6.3 Error Analysis

Table 4 summarises false positive and false negative counts for all three transformers on the test set.

MuRIL achieves the most favourable error profile (FN = 10, FP = 18), followed by BanglaBERT (FN = 11, FP = 21) and XLM-RoBERTa (FN = 14, FP = 23).

Both error directions reveal consistent failure patterns across all three models.

Model	Class	P	R	F1	FP	FN
MuRIL	Non-Rumour	0.9855	0.9743	0.9799	18	–
	Rumour	0.9746	0.9857	0.9801	–	10
BanglaBERT	Non-Rumour	0.9841	0.9700	0.9770	21	–
	Rumour	0.9704	0.9843	0.9773	–	11
XLM-RoBERTa	Non-Rumour	0.9797	0.9671	0.9734	23	–
	Rumour	0.9676	0.9800	0.9737	–	14

Table 4: Per-class results for transformer models ($n = 1{,}400$).

False Negatives. A consistent subset of rumours is missed by all three models simultaneously.

Two categories dominate.

First, *meta-referential content*: headlines from Rumor Scanner and Jachai describing their own detection statistics and organisational activities. These items carry the linguistic register of credible investigative journalism, include named outlets, and lie in the Marker Signal Void (WMS ≤ 0).

Second, *neutral-register impersonation*: false announcements of government welfare schemes written in bureaucratic language that activate no positive marker and are surface-identical to authentic public service announcements.

Both patterns expose a structural limitation: once a rumour adopts the register of institutional or fact-checking discourse, neither marker-based nor distributional approaches can reliably flag it (Rashkin et al., 2017).

False Positives. The dominant false-positive pattern across all three models is *legitimate attributed speech* activating the `claim_verb` marker ($w = 0.171$), the second-highest-weight feature.

Political and international breaking-news headlines quoting named parties are often surface-identical to rumour headlines that quote false claims.

A second cluster comprises celebrity and entertainment headlines with dramatic vocabulary that simultaneously activates multiple lower-weight markers.

These findings align with prior observations that epistemic markers often misfire on legitimate attributed political speech (Pérez-Rosas et al., 2018).

7 Conclusion

We presented **MisInfo-11K**, the first domain-stratified Bangla rumour corpus (11,198 headlines, 7 domains), with a comprehensive eight-model benchmark spanning traditional and transformer architectures. We validated 14 Bangla-specific linguistic markers (all $p < 0.0001$), establishing domain-specific deception fingerprints — authority mimicry (political/international), emotional provocation (religious), and fear amplification (health) — with a +0.57 Macro-F1 gain from marker augmentation confirming independent discriminative value. The Marker Signal Void analysis establishes a theoretical detection ceiling of Macro-F1 = 0.9843; MuRIL achieves 0.9800 (ROC-AUC = 0.9971), with transformers recovering 8–12 of the 22 void rumours through contextual representations. Zero-shot evaluation on BanFakeNews (Hossain et al., 2020) confirms that rumour detection requires a dedicated headline-level resource. Future work will pursue propagation-graph integration (Bian et al., 2020), multimodal evidence fusion, and cross-lingual transfer to other under-resourced South Asian languages.

Limitations

Domain and Temporal Scope. MisInfo-11K covers headlines from 2024–2026; models may not generalise to earlier periods or future data as misinformation strategies evolve.

Headline-Only Input. All models operate on headline text truncated to 128 tokens. The zero-shot collapse on BanFakeNews (Macro-F1 = 0.4728) confirms that article-level discriminative signals — metadata, propagation context, and multimedia — are absent from the current framework.

Formal Written Bangla. The corpus covers formal registers only. Informal social-media language — dialectal variation, code-switching, WhatsApp forwards — is not represented, which may limit robustness on real-world misinformation content.

Taxonomy Reliability. Domain labels were assigned programmatically; only a 200-instance subset received human IAA validation ($\kappa = 0.746$). Domain-level findings should be interpreted accordingly.

Single-Language Scope. The marker framework is validated on Bangla only. Transfer to related South Asian languages (Hindi, Assamese, Sylheti) remains an open empirical question.

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